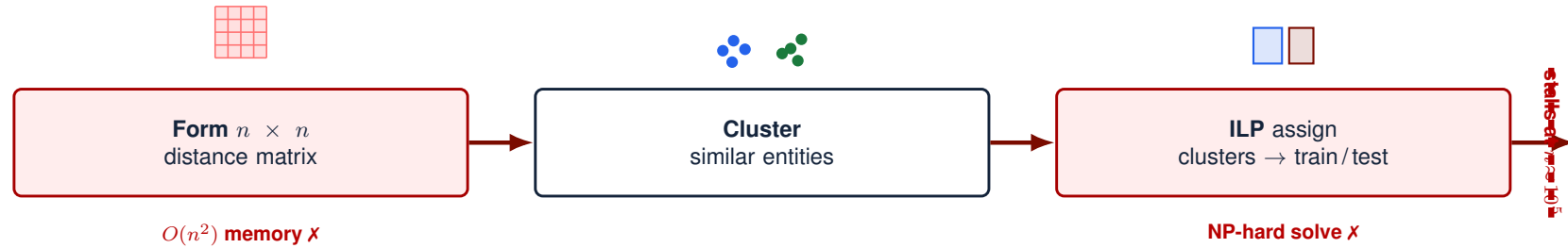


The baseline: DataSAIL

DataSAIL (*Data Splitting Against Information Leakage*) is the standard leakage-aware splitter and our closest baseline. It targets the *same* cross-split similarity — but solves it **exactly**, and every step is $O(n^2)$ or worse:



Faithful but **unscalable**: it must *materialize* the whole $n \times n$ similarity and solve a combinatorial program — **90 min** at 100k, **out-of-memory** by 300k. **This is the gap we factorize away.**